MATHEMATICS AND STATISTICS – 1

Chapter 3 | Arithmetic Progression | 2 hrs | 40 Marks

Q.1. (A) Choose the correct alternative. 5 × 1 = [05]

1) Which of the following is an A.P.?

a) 2, 4, 8, 16,... b) 3, 6, 9, 12,... c) 1, 4, 9, 16,... d) 2, 3, 5, 8,...

2) The common difference of the A.P. 7, 11, 15, 19,... is

a) 2 b) 3 c) 4 d) 5

3) The first term of the A.P. 5, 8, 11, 14,... is

a) 3 b) 5 c) 8 d) 11

4) The 10th term of the A.P. 2, 5, 8, 11,... is

a) 27 b) 29 c) 30 d) 32

5) The sum of first 10 natural numbers is

a) 45 b) 50 c) 55 d) 60

Q.1. (B) Solve the following. 5 × 1 = [05]

1) State whether $(m + 2)(m - 5) = 0$ is a quadratic equation.

2) Write $2x = 10 - x^2$ in standard form and identify a, b and c.

3) Verify whether $x = 2$ is a root of $x^2 - 5x + 6 = 0$.

4) Find k if $x = 3$ is a root of $kx^2 - 10x + 3 = 0$.

5) Find the value of k if the roots of $x^2 + kx + 16 = 0$ are equal.

Q.2. (A) Complete the following activities. (Attempt any 2) 2 × 2 = [04]

1) Complete the activity to solve

For the A.P. 1, 8, 15, 22,...

$a = \square$

$d = \square$

$t_4 = a + (4 - 1)d$

$= \square + 3(\square)$

$= \square$

2) Complete the activity to find the nature of roots of

For the A.P. 3, 7, 11, 15,...

$a = \square$

$d = \square$

$t_{10} = a + (10 - 1)d$

$= \square + 9(\square)$

$= \square$.

3) Complete the activity to form a quadratic equation whose roots are 4 and 7.

Find S_5 for the A.P. 2, 5, 8, 11, 14,...

$a = \square$

$d = \square$

$S_5 = 5/2\ [2a + (5 - 1)d]$

$= 5/2\ [\square + \square]$

$= \square$

Q.2. (B) Solve the following simultaneous equations. (Attempt any 3) $3 \times 2 = [06]$

1) Find the 15th term of the A.P. 8, 12, 16, 20,...

2) Find the first term of an A.P. whose common difference is 5 and 8th term is 39.

3) Find the common difference if $a = 7$ and $t_{12} = 40$.

4) Find S_{20} of the A.P. 3, 6, 9, 12,...

Q.3. (A) Complete the following activity. (solve any 1) $1 \times 3 = [03]$

1) Complete the activity.

Find the 25th term of the A.P. 6, 9, 12, 15,...
$a = \square$
$d = \square$
$t_{25} = a + (25 - 1)d$
$= \square + 24(\square)$
$= \square$

OR

2) Complete the activity.

Find S_{10} of the A.P. 1, 3, 5, 7,...
$a = \square$
$d = \square$
$S_{10} = 10/2\ [2a + (10 - 1)d]$
$= \square\ [\square + \square]$
$= \square$

(B) Solve the following word problems. (Attempt any 2) $2 \times 3 = [06]$

1) Find the sum of first 30 natural numbers.
2) Find the sum of first 20 odd natural numbers.
3) The first term of an A.P. is 10 and common difference is 3. Find the sum of first 25 terms.
4) How many terms of the A.P. 7, 10, 13, 16,... are needed to give a sum of 330?

Q.4. Solve the following. (Attempt any 2) $2 \times 4 = [08]$

1) The sum of the first n terms of an A.P. is given by $S_n = 2n^2 + 5n$

Find: i) The first term. ii) The common difference. Iii) The 15th term.

2) A labourer was appointed on a contract for 30 days on the condition that he would receive ₹250 on the first day and thereafter his daily wage would increase by ₹20 every day.

Find:

- His wage on the 30th day.
- The total amount earned by him during the 30 days.

3) A cinema hall has 30 seats in the first row. Every successive row has 2 more seats than the previous row. If there are 25 rows, find:

- The number of seats in the last row.
- The total seating capacity of the cinema hall.

Q.5. Solve the following. (Attempt any 1) $1 \times 3 = [03]$

1) On 1st January, Sanika saves ₹10, ₹11 on the second day, ₹12 on the third day and so on. Find her total saving after 365 days.

2) A person deposits ₹1000 in the first month, ₹1200 in the second month, ₹1400 in the third month and so on. Find the total amount deposited in 12 months.

✦ All The BEST! ✦